

Options Trading LLM - Training Analysis

1. Current Model Performance

Metric	Value	Status	Notes
Loss	0.402	Converged	Flat since epoch 2.5
Accuracy	89.2%	Good	On synthetic test set
Training Data	19K examples	Curated	8 strategies balanced
Cost	\$8.15 / \$15	Under budget	43% budget remaining
Epochs	3.0 / 4.0	Stopped early	Loss plateau detected
LoRA Config	r=16, alpha=32	Standard	13.6M trainable params

2. Verdict: Deploy Now, Save Modal Compute for Later

The loss plateau at 0.402 means the model has learned everything it can from the synthetic training data. More epochs on the same data will not improve performance. The real test is paper trading, not training metrics.

3. Why More Training Now Wont Help

Issue	Detail
Loss already flat	0.402 since step 600+, no improvement possible
Synthetic data bias	82.6% of examples are BUY actions
No real market context	Synthetic Greeks and price data vs real Alpaca feeds
Accuracy not profit	89.2% on synthetic data does not guarantee profitability

4. What the \$20 Modal Compute Is Best Used For

Use Case	Value	Explanation
More epochs (same data)	Zero	Loss already flat, no improvement
Higher LoRA rank	Marginal	Capacity is not the bottleneck
Train on real data	High	But we do not have this data yet
Retrain after feedback	Best	Target actual weaknesses from paper trading

5. Recommended Path Forward

- Step 1: Deploy the Modal inference endpoint and start paper trading immediately.
- Step 2: Collect real feedback for 1-2 weeks. Track which trades made or lost money.
- Step 3: Use the \$20 Modal compute for a second training round on real market data, targeting the specific weaknesses identified in paper trading.

6. Training Timeline

Step	Train Loss	Eval Loss	Accuracy	Epoch
10	1.198	-	78.3%	0.03
50	0.609	0.575	85.8%	0.17
100	0.472	0.429	88.1%	0.34

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300	0.421	0.424	88.8%	1.01
600	0.405	0.406	89.1%	2.02
900	0.402	0.404	89.2%	3.03

7. Conclusion

The Phase 1 training on Lightning RTXP 6000 is complete. The LoRA adapter is trained, uploaded to HuggingFace Hub, and ready for deployment. The \$20 Modal compute budget should be reserved for a targeted second training round after real-world paper trading feedback identifies specific model weaknesses. Deploying now costs nothing and provides the most valuable data for future improvements.